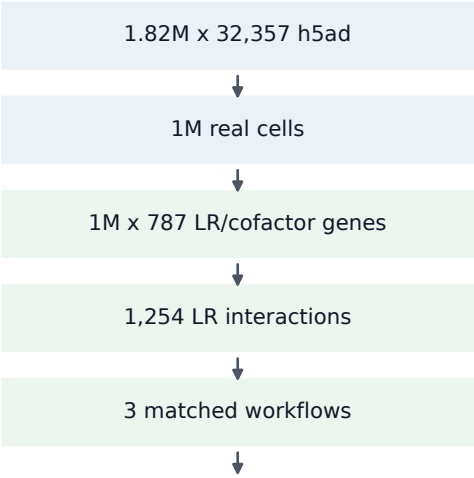
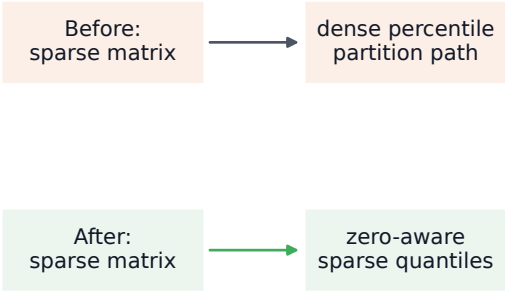


A Benchmark design



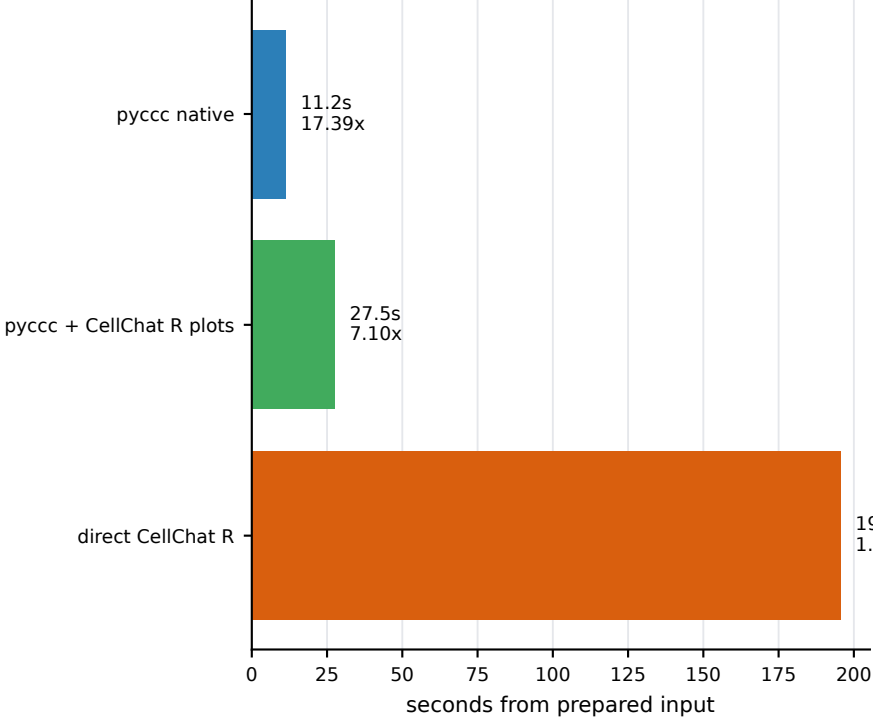
Strict 4-core affinity; prepared 1M input; regular mode only.

D Sparse tri-mean optimization

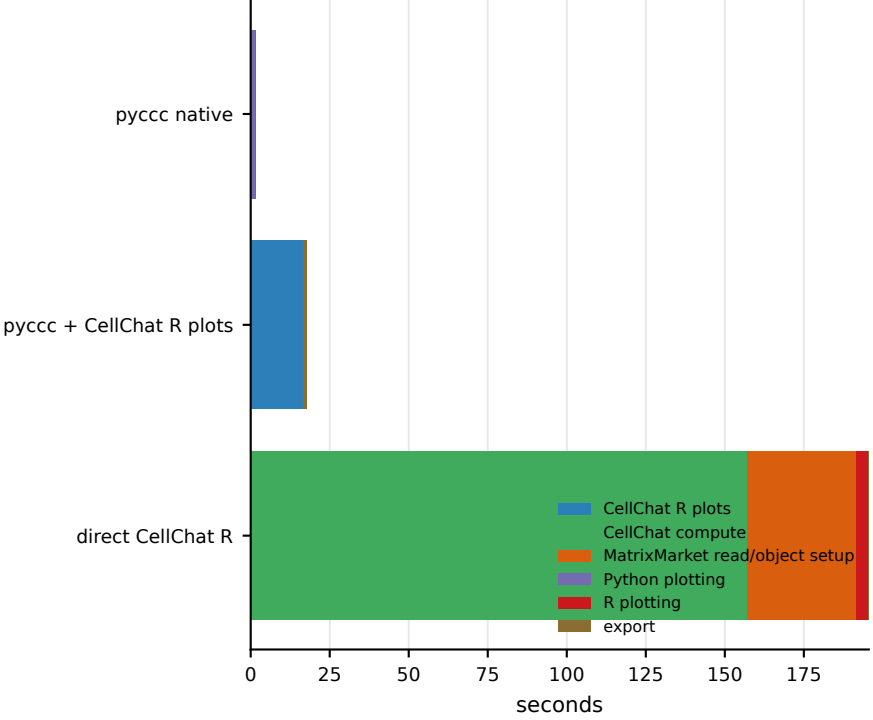


The speedup comes from exact sparse aggregation, not high worker count.

B Runtime



C Time decomposition



F Timing policy

Counted: reading the prepared 1M pyccc h5ad; direct CellChat reading MatrixMarket inside R; analysis and matched visualizations.

Not counted: sampling from the full 1.82M h5ad and LR filtering/preparation.

Caption must state: prepared 1M input, strict 4-core affinity, regular mode only.

E Peak CPU

